23

Blessed G/Michael Catholic Secondary School

1st Semester Mathematics Test Two for Grade 12th

Nov , 2018 E.C

Time Allowed ____1:30__min'

Name Hanna Ayalew Section 12th C No 24.

I. Choose the best answer among the given alternatives.

A 1, The rat of change of (3, 4) and (6, 5) is:

A.1/3 B, 4/3 C, 3 D, 5/4

B 2, The average rate of change of a function f(x)= 3x-4 between x=4 and x= 6 is :

A, 4 B, 3 C, 14 D, 8

A 3, The instantaneous rate of change of $f(x) = \sqrt{x}$ at x=1 is:

A, ½ B, ¾ C, does not exist D, 4

C 4, The instantaneous rate of change of $f(x) = 3x^4 + 2x^2 + 4$ at x=0 is :

A, 4 B, 2 C, 0 D, 1

C 5, The slope of the secant line to the graph of the function $f(x) = 3x^2-4x+6$:[1,2] is :

A, 4 B, 6 C, 5 D, 3

D 6, The gradient of the tangent line to the graph of $f(x) = \frac{1}{x}$ at (4, 1/4) is :

A, $\frac{1}{4}$ B, $\frac{3}{4}$ C, $\frac{2}{3}$ D , $\frac{-1}{16}$

A 7, The derivative of $f(x) = \sqrt{3x+4}$ at x=0 is:

A, $\frac{3}{4}$ B, $\frac{4}{3}$ C, $\frac{1}{2}$ D, 3

B 8, Let $y=1/(2x^5)$, then $\frac{dy}{dx}$ of at x=1 is :

A, $\frac{1}{2}$ B, $\frac{-5}{2}$ C, $\frac{4}{5}$ D, $\frac{3}{2}$

D 9, The derivative of $f(x) = 3x^{-4}$ is---:

A , $\frac{10}{x^5}$ B, $\frac{12}{x^4}$ C, $\frac{10}{x^3}$ D, $\frac{-12}{x^5}$

B 10, The derivative of $f(x) = \sqrt{x} - \frac{1}{2\sqrt{x}}$ at =0 is :

A, 1 B, undefined C , 0 D, 4

D 11, Let $y=3x^2+\frac{1}{2x^2}-6$, then the value of $\frac{dy}{dx}$ at x=1

A, 4 B, 1 C, 6 D, 5

A 12, The derivative of $f(x) = (x^2-1)(x^3+2x)$ is :

A, $5x^4+3x^2-2$ B, $5x^4+3x^2+2$ C, $4x^3+3x^2-4$ D, $6x^4+2x+4$

C 13, Let $y=\frac{4x^3}{2x3+4x}$, $\frac{dy}{dx}$ at x=1 is :

A, $\frac{7}{3}$ B, $\frac{9}{4}$ C, $\frac{8}{9}$ D, ¾

C 14, The derivative of $y=\frac{x3+4x}{x^2}$ is ;

A, 1 B, $\frac{1}{x}$ C,4x+1 D, $\frac{1}{x^2}$

15, Which one of the following is the value of $\frac{dy}{dx}$ at x=2 for f(x) = $(\sqrt{x^3+4x})^5$ is

A, $800\sqrt{2}$ B, $40\sqrt{2}$ C, $4\sqrt{2}$ D, $2\sqrt{2}$

16. The maximum value of f(x)=x(x-6) on the interval [-4 ,4] is :

A, f(-4) B, f(3) c, f(0) D, f(1)

17. The slope of the graph of f(x)= $\frac{x+1}{x+2}$ at x=0 is :

A, 2/5 B, 1/3 C , 1/ 4 D, 4

18. for what value of a f(x)= $\frac{x}{x^2+a}$ will have extreme value at x= 2 is:

A, 4 B ,5 C, 2 D, 6

19 Given that g(8)=1 , $g'(8)=0$,f(8)=-1 and $f'(8)=2$ for F(x)= $\frac{3x-1}{g(x)}$, the value of $F'(8)$ is:

A, 8 B, 5 c, 3 D, 6

20. If f (2) = 3 , g(2)=-2 , $g'(2)=-1$ and $f'(2)=5$ then which one of the following is the value F(x)= $\sqrt{f(x)}$ is :

A, $\frac{5\sqrt{3}}{6}$ B, 2/5 C, 4/3 D, 4

II. Short answer

16, Let y= $\frac{(\sqrt{3x^2+4x})}{3-4x}$, then find $\frac{dy}{dx}$ at x=1

17, Let f(x) = $\frac{2x^2+4x}{x+4}$,then the derivative of f(x) is ?

18 , Find the critical number of $f(x) = 4x^3 - 4x^2 + 6$ is ------

19, find the intervals on which the functions are increasing and decreasing for

$f(x)=(x^2+6x+9)(x^2-2x+1)$

20, find the absolute maximum and minimum values of $f(x)=3x^4-12x^3-20x^2+2$